

# STEP 11 — PAKISTAN PENETRATION VINTAGE: SOURCE RE-AUDIT AND FACADE SENSITIVITY

## Referee context

The panel's largest facade quantities ( $R_{\text{facade}} = 5.28$ ,  $\Phi = 0.413$ ) rest on its weakest input: Pakistan's internet penetration is a 2020 ITU carry-forward (18.93%), while the ITU 2024 release reports approximately 57%. A discontinuity of that size is produced by survey-methodology revision, not organic growth — and when methodology changes, ITU typically back-revises the whole series. A referee will note the asymmetry with the China correction (§5.4) and can check the DataHub in minutes. This step first re-audits the source; if the carry-forward survives, it then converts the vulnerability into two robustness results and an interval presentation.

## Trial-run findings (pending your verification — script attached)

Under a 57% nowcast:  $R_{\text{facade}} = 1.754$  and  $\Phi = 0.2189$ . The paper's formal criterion (§3.5: intersection of the four largest facade ratios and the four largest stranded capacities) still returns exactly {Bangladesh, Nepal, Pakistan} — Pakistan is fourth on both quantities, with a  $\Phi$  margin over Burkina Faso of only 0.003. Recomputing the composite with the nowcast gives Pakistan HDI-N = 0.363, still between Nepal (0.418) and Burkina Faso (0.272): rank 18 is unchanged, so the vintage issue touches the facade quantities only and no ranking claim in Table 4. All three findings are trial-run values; none enters the paper until you reproduce them from the audited files.

## Tasks

| # | Task                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Standard                                                                                                                      |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>Source re-audit (GATE).</b> Pull ITU DataHub series i99H (Individuals using the Internet) for Pakistan as retrieved today. Record every year 2019–2024 displayed with its exact value, and whether the series has been back-revised since our original pull. Report an explicit branch call: do within-window (2021–2023) values now exist — yes or no?                                                                                                                                                                                                                                        | CSV/printout with retrieval date and source string; unambiguous yes/no branch call before any other task starts               |
| 2 | <b>Branch A (within-window value now exists).</b> Recompute Pakistan per the China precedent: penetration $\rightarrow F_{\text{egov}}, T_{\text{egov}}^{\text{adj}}, R_{\text{facade}}, \Phi, I_{\text{digital}}, \text{HDI-N}$ . Deliver revised CSV rows and a complete touch-list of every table, figure, and passage affected (Tables 1–4; Figures 2, 3, 5; abstract; §§1, 2.2, 3.2, 3.5, 5.1, 5.5, 6). Do not draft prose — replacement passages will be supplied.                                                                                                                          | Recomputation reproduced independently from audited source files; touch-list complete (checked against a grep for "Pakistan") |
| 3 | <b>Branch B (carry-forward stands).</b> Verify the three trial findings with the attached script against the audited files: (i) nowcast $R$ and $\Phi$ ; (ii) intersection criterion returns exactly the trio at both interval ends, $\Phi$ margin 0.003; (iii) HDI-N 0.363, rank 18 invariant. Also verify the exact ITU 2024 figure and its vintage — is 57% the release value? Record the precise number; it sets the interval's lower bound.                                                                                                                                                  | All three reproduced to 3 d.p.; exact 2024 value documented with source string and access date                                |
| 4 | <b>LaTeX and figure integration (Branch B).</b> Integrate supplied passages: emphasis sweep to Bangladesh–Nepal–Pakistan order (§1 twice; §3.2 including deletion of the erroneous "by an order of magnitude" claim; §3.5 restructure with the evidential-status statement moved up front; §5.1; §6); interval presentation $R \in [1.75, 5.28]$ , $\Phi \in [0.219, 0.413]$ in §3.5 with Table 2 caption note; new §4.7 (both invariance results); one §5.4 sentence naming the China asymmetry. Figure 2: caption sentence now; whisker marker for the nowcast position only if Ma'am approves. | Compiles clean; grep for "5.28" shows no remaining Pakistan-first framing outside Table 2 and the interval statements         |

## Watch-outs

**57% is a 2024 release value used as a nowcast** — never present it as a 2023 observation. **Do not overclaim the intersection survival:** the  $\Phi$  margin at the lower bound is 0.003; the drafted text says "marginal" and must stay that way.

**Branch B does not delete** the §2.2 carry-forward disclosure; **Branch A deletes it** and every upper-bound phrase — sweep the abstract, §3.5, §5.5, and §6 for orphans. Table 2 keeps its  $\Phi$  sort in both branches; the emphasis reordering is prose and captions only.

## Sequencing

Task 1 gates everything and takes under an hour: run it first and report the branch call before touching tasks 2–4. Branch A supersedes tasks 3–4 and triggers a bounded rerun (Pakistan rows only). Step 11 is independent of Step 9 (conversion indicator) and Step 10 (disparity reframe) and runs in parallel with both; all three land in the same revision.